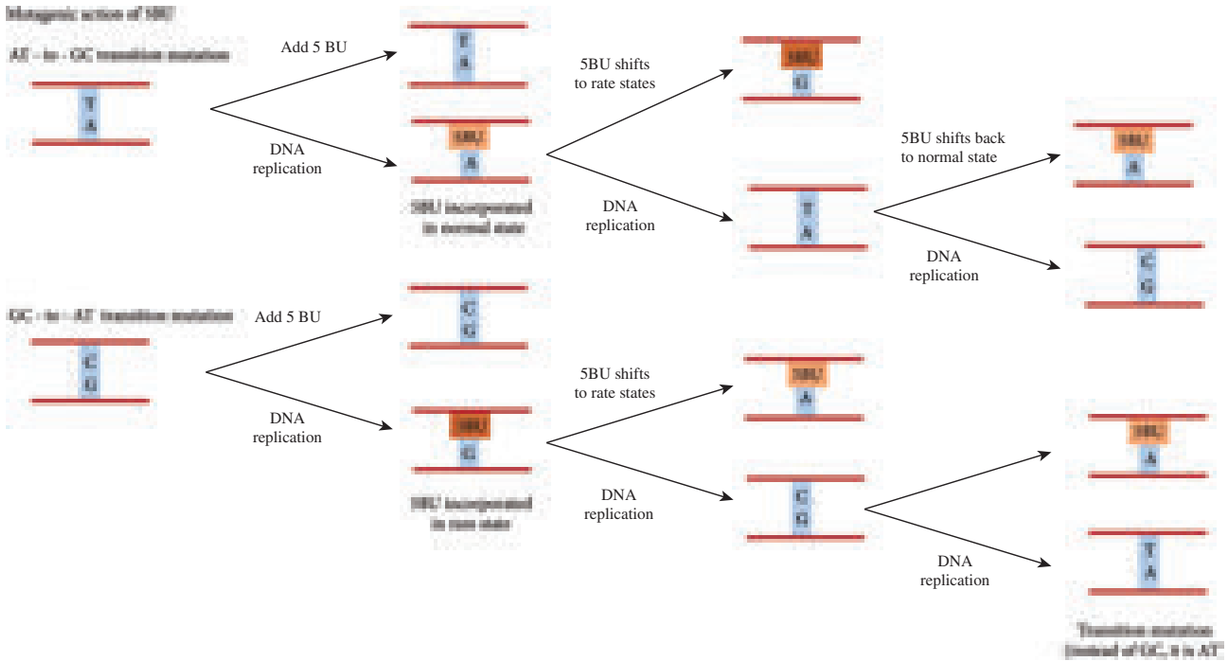


Figure 12.11: Mutagenic effects of 5-BU

adenine. When cytosine is treated with nitrous acid it removes amino group from cytosine which changes to uracil resulting in transition mutation (Figure 12.12a).

2. Hydroxylamine (NH_2 OH) is a hydroxylating mutagen that react specifically with cytosine, modifying it by adding a hydroxyl group (OH) so that it can pair solely with adenine instead of with guanine. Mutations induced by hydroxylamine can only be CG to TA transitions (Figure 12.12b).
3. Alkylating agents like methymethane sulfonate (MMS) introduces alkyl groups onto the bases at a number of location. For example after treatment with MMS, some guanines are methylated at 6 – oxygen to produce O6 – methyl guanine. The methylated guanine pairs with thymine rather than cytosine giving G C to AT transitions (Figure 12.12c).

Original base	Mutagen	Modified base Pairing partner	Predicted transition
Cytosine	Nitrous acid (H_2NO) ⟶	Uracil ═══ Adenine	CG TA
Cytosine	Hydroxylamine (NH_2OH) ⟶	Hydroxyl amino cystosine ═══ Adenine	CG TA
Guanine	Methylmethane sulfonate(MMS) (alkylating agent) ⟶	O-Methylguanine ═══ Thymine	CG TA

Figure 12.12: Action of three base modifying agents.

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